

SysML Model

Autonomous Vehicle – D2-D4 Project

Systems Engineering D2-D4 Project

This document contains the SysML model for a small autonomous vehicle built as a university D2-D4 project. The vehicle follows a black line on a track and reacts to obstacles using ultrasonic sensors. It is built from an Arduino, a motor controller, two DC motors, two IR sensors, two ultrasonic sensors, a plywood chassis, and a battery. The model covers requirements, structure, and behaviour using seven SysML diagrams.

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1 Requirement Diagram

Requirement Diagram - Autonomous Vehicle (D2-D4 Project)

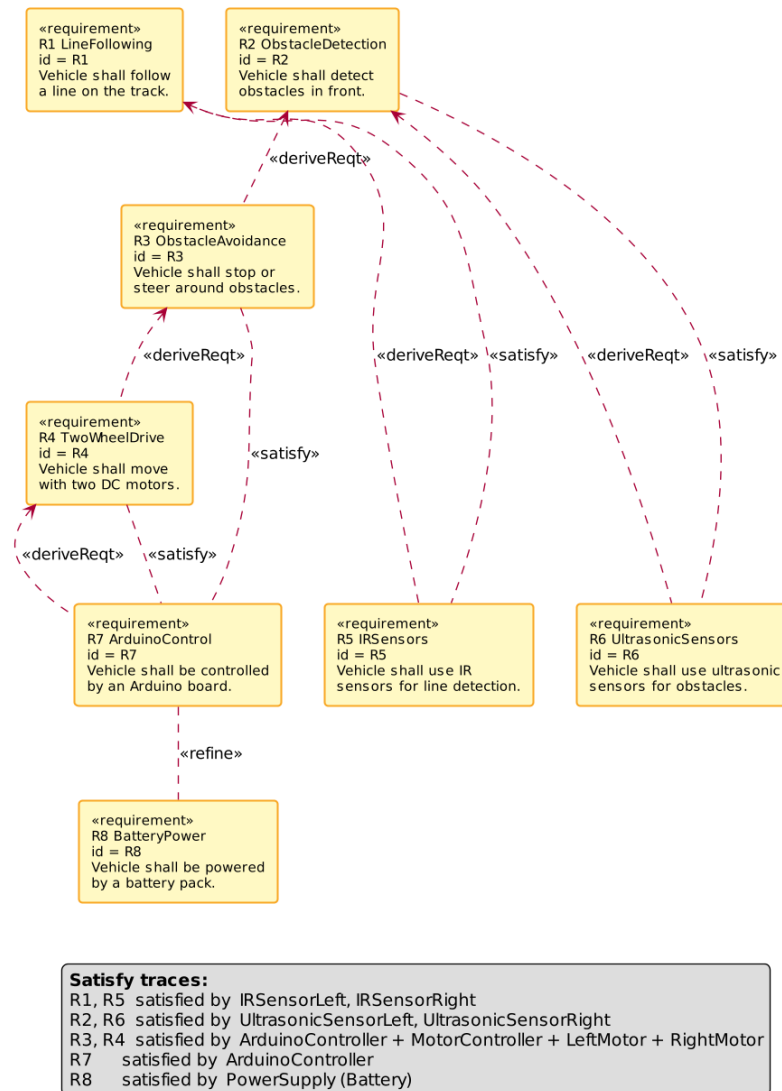


Figure 1. Requirement Diagram.

Explanation. This diagram shows the eight requirements for the D2-D4 project (R9 removed). R1 (line following) and R2 (obstacle detection) are the two main goals. <> arrows show that R5 (IR sensors) comes from R1 and R6 (ultrasonic sensors) from R2. R3 (obstacle avoidance) is derived from R2, and R4 (two-wheel drive) from R3. <> arrows show which block satisfies which requirement. R7 (Arduino control) refines R8 (battery power) because the Arduino depends on the battery.

2 Package Diagram

Package Diagram - Autonomous Vehicle Model

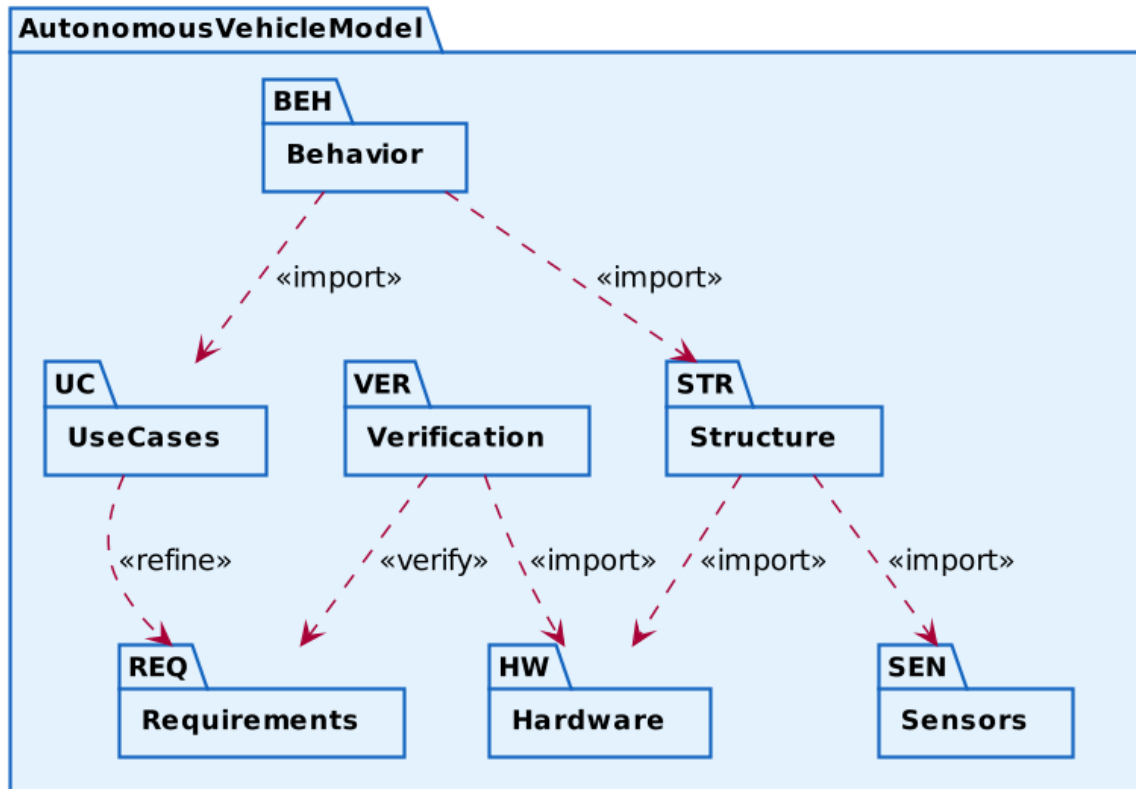


Figure 2. Package Diagram.

Explanation. The package diagram organises the model into seven simple packages. Requirements is the starting point. UseCases refines it. Structure and Hardware describe the physical build. Behavior describes how it works. Verification checks that the requirements are met. The arrows show which packages depend on which.

3 Use Case Diagram

Use Case Diagram - Autonomous Vehicle

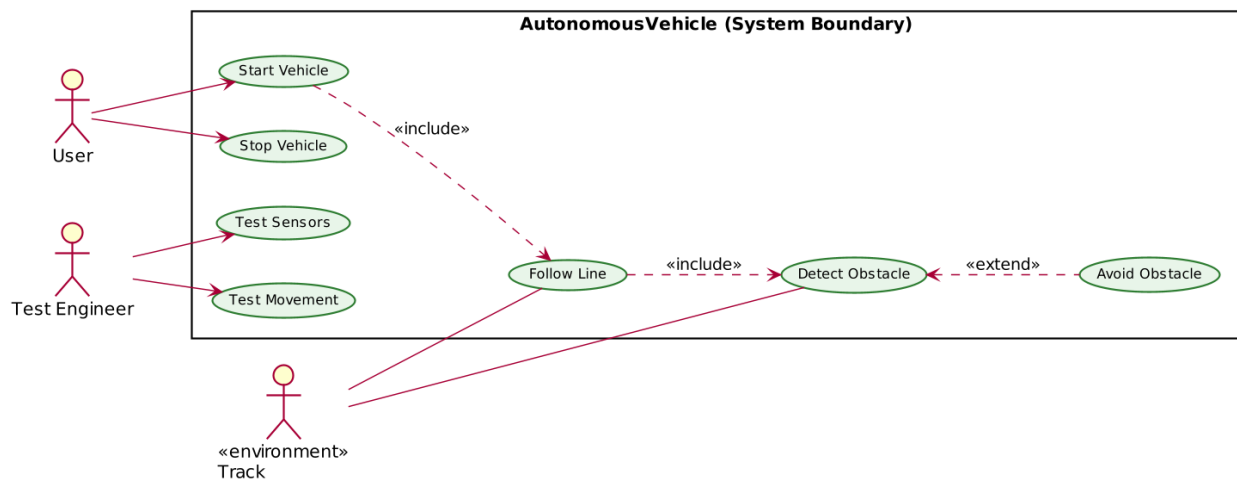


Figure 3. Use Case Diagram.

Explanation. The use case diagram shows what the system does and who interacts with it. The User starts and stops the vehicle. The Test Engineer runs sensor and movement tests. The Track (environment) is involved in line following and obstacle detection. <> means the use case is always called; <> means it only happens when needed (e.g. avoidance only when an obstacle appears).

4 Block Definition Diagram (BDD)

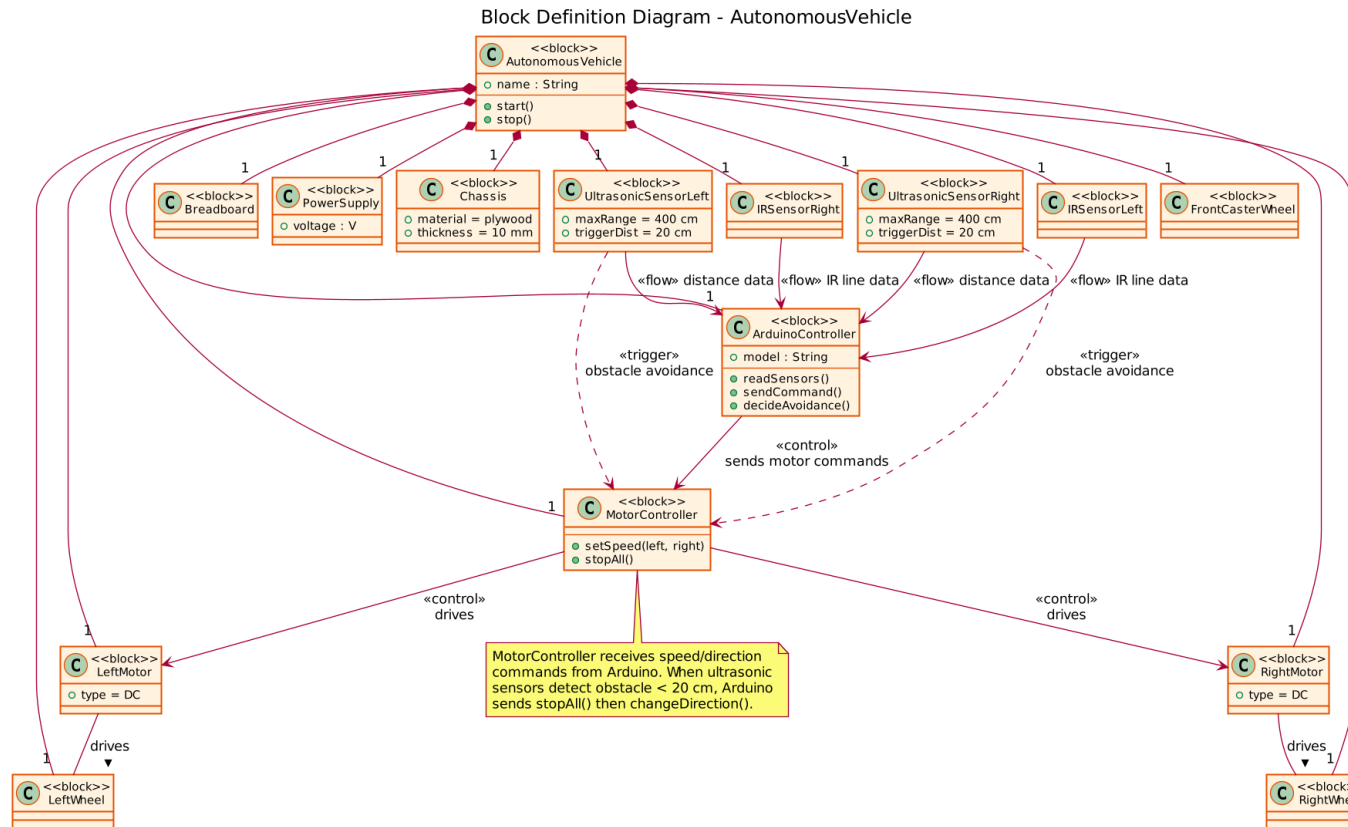


Figure 4. Block Definition Diagram (BDD).

Explanation. The BDD shows all hardware blocks and their key relationships. Filled diamonds (composition) show the AutonomousVehicle owns each part. Ultrasonic sensors send distance data to the Arduino (<>). When distance < 20 cm, the Arduino sends stopAll() then changeDirection() to the MotorController via <> / <> relations. The MotorController drives LeftMotor and RightMotor independently via PWM. Each motor drives its wheel mechanically. The yellow note explains this avoidance chain.

5 Internal Block Diagram (IBD)

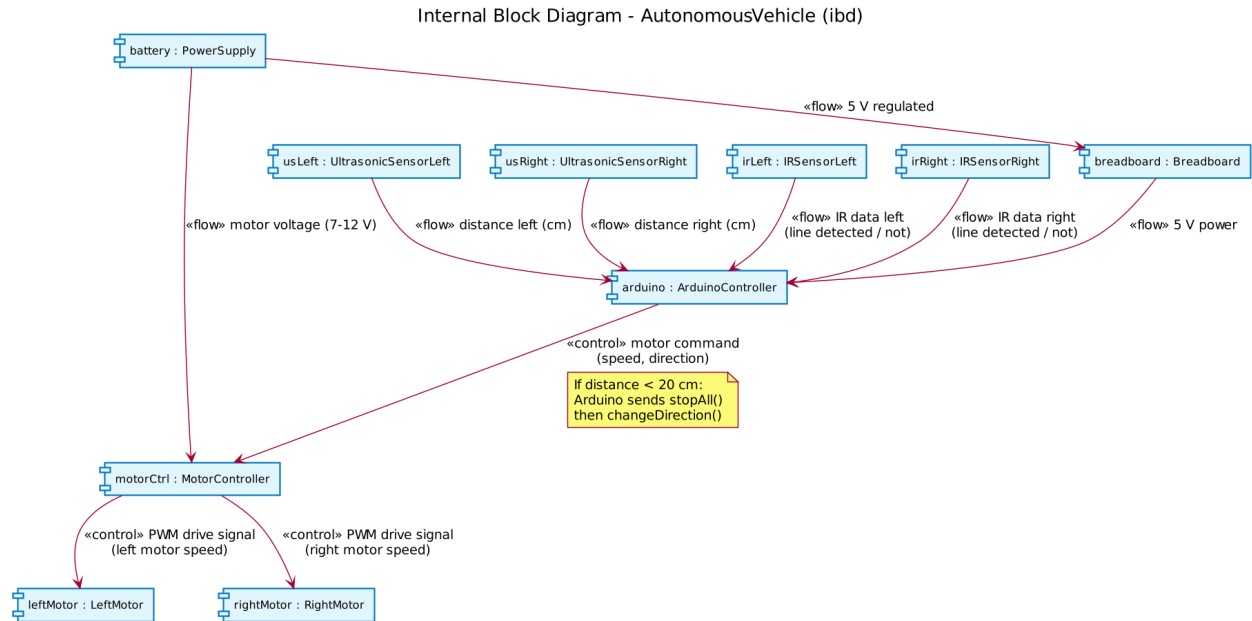


Figure 5. Internal Block Diagram (IBD).

Explanation. The IBD shows how the parts are wired and how data flows at runtime. The battery supplies 5 V (via breadboard) to the Arduino and higher motor voltage (7-12 V) directly to the MotorController. IR sensors send line detection signals (line found / not found) to the Arduino every cycle. Ultrasonic sensors send their measured distance in cm to the Arduino. The Arduino processes all sensor data and sends a motor command (speed + direction) to the MotorController. The sticky note on this link highlights the obstacle avoidance logic: if distance < 20 cm, Arduino calls `stopAll()` then `changeDirection()`. The MotorController drives the left and right motors independently using PWM signals, allowing the vehicle to steer by varying the speed of each motor.

6 Activity Diagram

Activity Diagram - Autonomous Vehicle Behavior

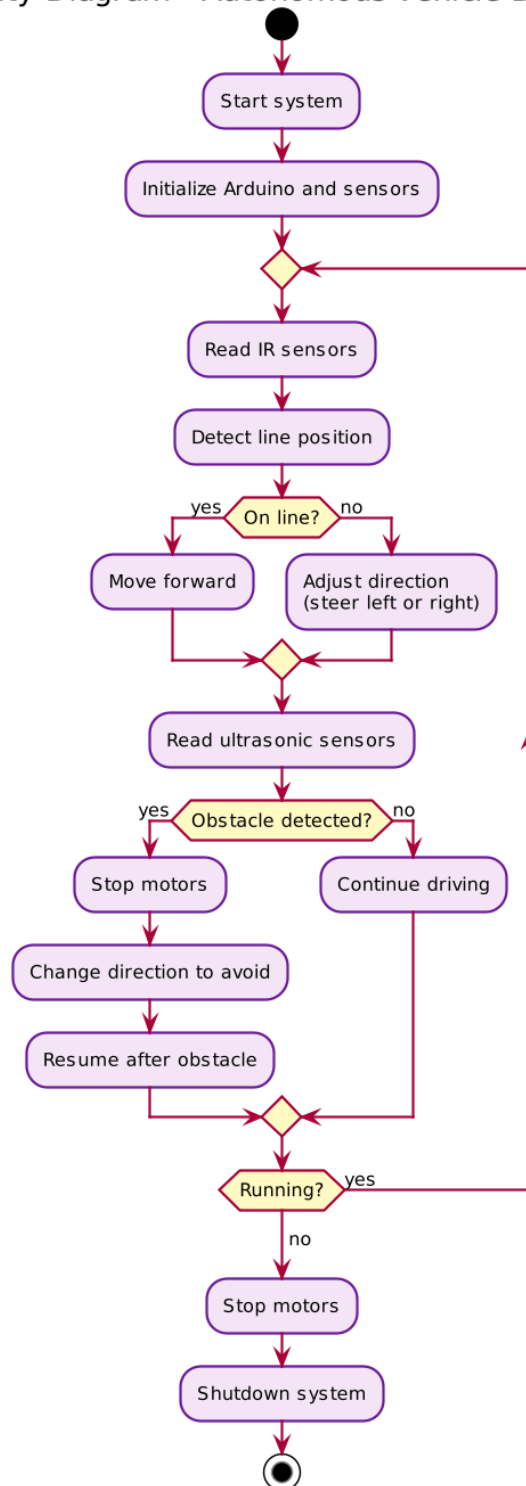


Figure 6. Activity Diagram.

Explanation. The activity diagram describes the main behaviour loop. After startup the Arduino reads the IR sensors and decides whether to go straight or steer. Then it reads the ultrasonic sensors. If an obstacle is close, it stops, changes direction, and resumes. This loop keeps running until the user stops the vehicle.

7 Sequence Diagram

Sequence Diagram - Line Following with Obstacle Reaction

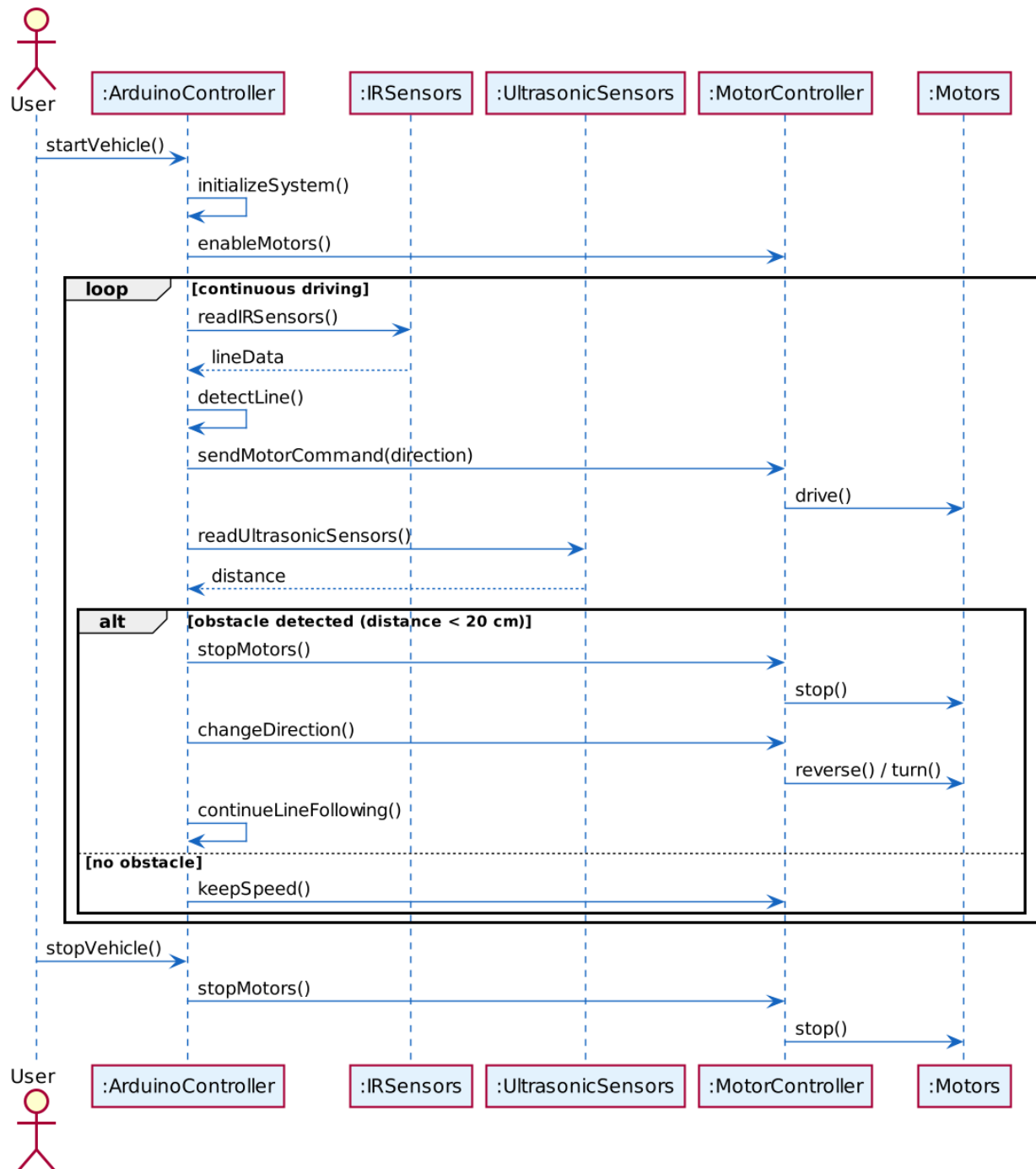


Figure 7. Sequence Diagram.

Explanation. The sequence diagram shows the message order for one driving cycle. The User starts the vehicle. The Arduino initialises and enables the motors. Inside the loop the Arduino reads IR and ultrasonic sensors every cycle. The alt fragment shows two cases: if an obstacle is detected, the motors stop and the direction changes; otherwise the vehicle keeps its speed. The User can stop the vehicle at any time.